

Phenix Validation

7 documentation pages

Generated from local Phenix documentation

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Model Validation with Crystallographic Data

validation.html

Why

Comprehensive Validation (X-ray) is a tool that can be used to evaluate the plausibility of a model, including its agreement with crystallographic X-ray or neutron data. Note that for cryo-EM data you should use instead Comprehensive Validation (cryo-EM).

You should validate an atomic model generated from crystallographic experiments by checking if it is consistent with priory knowledge about stereochemistry. This information includes covalent geometry, non-bonded interactions, and known distributions of side chain and main chain conformations in both proteins and nucleic acids. This validation should also extend beyond purely geometric measures to include local fit to electron density.

Validation is performed throughout the structure solution process, which consists of cycles of validation, manual or automated rebuilding, and automated refinement until a satisfactory model is obtained. Useful validation criteria depend on the resolution of the data.

How

In Phenix, model validation is provided in the comprehensive validation GUI based on the MolProbity tool. This software can be run as a standalone program or automatically as part of phenix.refine. Overall model statistics are presented in a summary chart with local scores reported as plots and as tables listing the outliers on each criterion. The model validation tasks are essentially identical to the MolProbity web service. The validation results are seamlessly integrated with the graphics programs Coot and PyMOL.

How to use the validation tools in the Phenix GUI: [Click here](#)

More information about MolProbity

Common issues

Outliers versus errors: In most structures, there will Ramachandran and side chain rotamer outliers. If the electron density and surrounding chemistry supports the outlier conformation, then it is most likely an outlier from prior expectations. In the absence of supporting density and/or conflicting local chemistry, it is probably an error that needs correction.

Related programs

- `phenix.model_vs_data`: This program assesses the overall quality of an atomic model with respect to the experimental data.

Structure Exploration with CaBLAM Training

`cablam_training.html`

Authors

`cablam_training`: Christopher Williams (Richardson Lab, Duke University)

Purpose

`cablam_training` is primarily a set of developer tools for exploring protein structure conformation. It provides tools for assessing protein geometry, with both traditional measures and with the idiosyncratic CaBLAM measures. It contains tools for identifying known and novel motifs based on hydrogen bonding, amino acid sequence, and other properties. And it contains tools for extracting motif and structure information into forms usable by CaBLAM validation for assessing low-resolution protein structures.

For information on `cablam_validate`, the main user-end interface for CaBLAM, please refer to the documentation for CaBLAM Validation.

The Short Version

Running "`phenix.cablam_training help=True`" on the commandline will provide fairly complete documentation. This document is intended to expand on and supplement that documentation, not replace it.

How CaBLAM Training Works

For each residue in a submitted structure file, `cablam_training` calculates several geometry measures selected by the user. It will print these measures for various user-defined selections. These selections may be residues of a certain type or may be residues that are members of a structural motif of interest.

Available Measures

Many standard measures are available through `cablam_training`. These include: - `rama=True` for Ramachandran dihedrals - `exrama=True` for expanded Ramachandran dihedrals, that is $\psi-1$ and $\phi+1$ in addition to ϕ and ψ - `tau=True` for the N-CA-C backbone angle - `omega=True` for the peptide bond dihedral (used to define if a residue is cis or trans)

CaBLAM measures

The CaBLAM system uses some idiosyncratic measures of protein geometry. - Two CA-defined pseudo dihedrals. For residue i , `CA_d_in` (aka μ in) is defined by the CA positions of $i-2, i-1, i, i+1$ and `CA_d_out` (aka μ out) is defined by the CA positions of $i-1, i, i+1, i+2$. - One dihedral that relates peptide plane orientation across the residue. For residue i , `CO_d_in` (aka ν) is defined by $O(i-1)$, a virtual atom in $i-1$, a virtual atom in i , and $O(i)$. The virtual atoms are defined at the perpendicular intersection of O and the CA-CA line through the residue. -The CA virtual angle. For residue i , `CA_a` is defined by the CA positions $i-1, i, i+1$. `cablam=True` will return these measures and is the recommended output selection for `cablam_training`.

Some other measures are available as artifacts of the initial training and measure selection process. `CO_d_out` is equivalent to `CO_d_in`, but is calculated using the current and succeeding residues. `CA_a_in` and `CA_a_out` are the virtual angles for the preceding and succeeding residues. None of these measures are

recommended as more than curiosities.

Quality Control

Quality filtering your data is extremely important. cablam_training provided a few tools for this at the residue level. Not that because the CaBLAM measures for each residue are dependent on many of that residue's neighbors, pruning one residue with these controls will remove reporting for other residues in its proximity. This is judged worthwhile in the interests of quality, but users should be aware.

- `b_max=#.#` Sets a B-factor cutoff. Residues with any mainchain atom (N, CA, C, O) with a B higher than this number will be excluded from all calculations. Using `b_max=30.0` or stricter is strongly recommended unless you have good reason not to.
- `prune_alts=True/False` Residues with alternate conformations for any mainchain atom are excluded from all calculations. Useful if alternates are modeled inconsistently. Default behavior uses the first alt (and only the first alt) for each residue.
- `prune=restype1,restype2` Residues of the listed (3-letter) residue types are excluded from all calculations. `prune=GLY,PRO` will totally remove the effects of the "weird" residues from the reporting, for better or worse.

Other Output Options

The default printing is comma-separated text, with a column for the residue id and columns for each of the user-requested measures. A few modifications to this are possible: - `give_kin=True` instead of .csv format, prints a multidimensional point cloud in .kin format for use with the KiNG viewer. - `cis_or_trans=cis/trans/both` selects whether cis-peptides, trans-peptides, or both will be printed. The default is "both". `omega=True` must be used with this option. - `skip_types=restype1,restype2` Similar to `prune` in quality control, but less severe. Ignores these types for output, but not for calculation, so their neighbors can still be evaluated. - `include_types=restype1,restype2` Works with `skip_types` If only `include_types` is used, only the listed restypes will be printed. If `include_types` and `skip_types` are both used, then the types given to `include_types` will override those skipped by `skip_types`. Sequence relationships may be represented with underscores.

Motif Searching

Motif searching is another groups of output modes dedicated to looking at entire motifs, rather than individual residues. Motifs are primarily defined by hydrogen bonding pattern, but may also require specific residue types and cis/trans peptides.

Motif searching is activated by giving `probe_motifs=` one or more motifs to look for. Available motif names may be found with `list_motif=True`, or ambitious users may define their own. See `mmtbx/cablam/fingerprints/how_to.py` for details (`how_to.py` is not an actual script, it's a .py so that fancy editors will color the example code correctly.)

Because motifs are defined by hydrogen bonds, hydrogens and phenix.probe information, motif search will run `phenix.reduce` and `phenix.probe` for each file unless a precomputed .probe file is provided via `probe_path=path/to/probefiles`. If you expect to run motif searching many times, it may be worth precomputing these files. Run the following command on a `phenix.reduce'd` .pdb file to generate an appropriate file: - `phenix.probe -u -condense -self -mc -NOVDWOUT -NOCLASHOUT ALL filename.pdb > filename.probe` if you need sidechain H-bonding information - `phenix.probe -u -condense -self -mc -NOVDWOUT -NOCLASHOUT MC filename.pdb > filename.probe` if you only need mainchain H-bonding

Motif Search Outputs

These outputs generate automatically-named files in the working directory. These files may overwrite each other if the program is called multiple times.

- `probe_mode=kin` returns an automatically-named `.kin` file for each member residue in each motif. The `kins` are high-dimensional dotlists containing the measures specified in the commandline (see above for options) for each residue that falls in the specified place in the motif. Recommended for finding motifs of interest within large filesets.
- `probe_mode=instance` returns an automatically-named vectorlist kinemage file for each motif of interest. Each `kin` is a high-dimensional vectorlist that shows the path of a multi-residue motif through the measures specified in the commandline. An alternative way to view the information available through `probe_mode=kin`
- `probe_mode=annotate` returns an automatically-named kinemage file for each `pdb` file. These `kins` are balllists that highlight the selected motifs of interest if appended to existing kinemages of the structures. Recommended to aid close inspection of motifs in single files of interest.

CaBLAM Validation in Phenix

`cablam_validation.html`

Authors

`cablam_validate`: Christopher Williams (Richardson Lab, Duke University)

Purpose

CaBLAM stands for C-Alpha Based Low-resolution Annotation Method. It is a system designed to use protein CA geometry to evaluate mainchain geometry and identify areas of probable secondary structure. CaBLAM is intended for use in low-resolution structures where compound errors or ambiguities in a model may make the results of highly-sensitive measures of protein conformation, such as Ramachandran analysis, difficult or impossible to interpret.

How CaBLAM works

For each residue, `cablam_validate` calculates several measures based on CA geometry. It uses these measures as coordinates for comparison against contours of expected protein behavior derived from a high-quality dataset (Top8000). Residues that fall outside these contours are considered outliers and are reported in validation feedback.

`cablam_validate` also compares local CA geometry measures which are robust at low resolution against contours of expected secondary structure behavior. Residues that fall within these secondary structure contours are reported as probable members of that secondary structure type.

Running CaBLAM Validation

CaBLAM validation is currently run from the commandline. It accepts a `.pdb` file (or a structure file that phenix finds similarly readable.)

`phenix.cablam_validate file.pdb output=[choice of: full_kin, kin, ca_kin, oneline, text, points]`

Outlier Cutoff

- `outlier_cutoff=0.05` is the default. This captures the majority of modeling errors, but also may misidentify as outliers real-but-rare conformations such as those in loops.
- `outlier_cutoff=0.01` is also recommended to identify severe errors.
- `outlier_cutoff=0.005` is recommended for use with `output=ca_kin`.

Several output modes are available.

Full Kin

`output=full_kin` gives the most comprehensive validation output, recommended for general use. It generates two automatically named files in the working directory: `file_cablam_multi.kin` and `file_cablam_multi.pdb`, then opens these files together in a phenix.king window. The `.kin` file contains CaBLAM markup for mild (0.05) and severe (0.01) CaBLAM outliers and for CA geometry outliers. The `.pdb` file contains the original model, plus HELIX and SHEET-style records for secondary structure elements identified by CaBLAM. The KiNG window

combines this feedback into a single, interactive format.

Kin

output=kin prints a markup kinemage to screen. This markup should be appended to a kinemage of the structure (such as a multi-criterion kinemage generated by phenix.kinemage). The markup consists of a set of colored lines drawn over each residue identified as an outlier. The lines follow the path of one of CaBLAM's geometric measures, a measure of the dihedral relationship between adjacent peptide planes. Among the CaBLAM measures, this is the dihedral most likely to be modeled incorrectly at low resolution.

CA Kin

output=ca_kin outlier_cutoff=0.005 prints a markup kinemage for C-Alpha geometry to screen. This markup should be appended to a kinemage of the structure (such as a multi-criterion kinemage generated by phenix.kinemage). CaBLAM is dependent on CA geometry, so it is necessary to know where that geometry is unreliable. The markup consists of red lines drawn along the CA Virtual Angle for each residue with a CA geometry outlier, but the virtual angle is not necessarily the angle at fault. Note that the outlier_cutoff must be set differently from its default to get meaningful results.

Online

output=online prints a single line summary of CaBLAM statistics for each of the submitted proteins. Sample output follows:

```
pdbid:residues:peptide_outlier_percent:peptide_bad_outlier_percent:ca_outlier_percent
3dnd:340:4.4:0.6:0.00
```

- pdbid identifies the input file
- residues gives the number of measurable residues assessed by CaBLAM (residues near termini and chain breaks are not assessable by CaBLAM)
- peptide_outlier_percent gives the percent of residues with "mild" outlier status. Higher than 5% may be cause for concern.
- peptide_bad_outlier_percent gives the percent of residues with "severe" outlier status. Higher than 1% may be cause for concern.
- ca_outlier_percent give the percent of residues with CA geometry outliers. Higher than 0.1% may be cause for concern.

Text

output=text is the default. This provides columnated, comma-separated records for each outlier residue. Sample output follows:

```
residue,contour_level,loose_alpha,regular_alpha,loose_beta,regular_beta,threeten
THR A 37 ,0.01437,0.00000,0.00000,0.02612,0.00000,0.00000
PHE A 54 ,0.01350,0.00027,0.00000,0.00109,0.00000,0.00000
ASN A 99 ,0.04785,0.00000,0.00000,0.12777,0.04921,0.00000
ASN A 115 ,0.04685,0.04231,0.00000,0.00011,0.00000,0.00000
GLY A 136 ,0.03689,0.00013,0.00000,0.00595,0.00000,0.00000
```

- residue: A full residue identifier for the outlier
- contour_level: The contour level for protein behavior at which the residue falls, in fraction form (max of 1.0). Lower values indicate a more severe outlier. 0.05 is the default cutoff for outliers

- loose_alpha/regular_alpha: contour levels for expected alpha helix behavior. Higher values indicate greater confidence that the residue should be modeled as alpha helix. loose_alpha is the most reliable indicator, and its default cutoff is 0.001
- loose_beta/regular_beta: contour levels for expected beta behavior. Higher values indicate greater confidence that the residue should be modeled as beta sheet. regular_beta is the most reliable indicator, and its default cutoff is 0.001
- threeten: contour levels for expected three-ten helix behavior. Higher values indicate greater confidence that the residue should be modeled as three-ten helix. Its default cutoff is 0.001

Points

output=points prints to screen a dotlist of points in kinemage format to produce a pointcloud of outlier residues in CaBLAM space. Primarily a developer tool.

Records

output=records prints to screen HELIX and SHEET-style records for secondary structure elements identified by CaBLAM. Formatting for these records is not yet totally consistent with PDB standard, but will be brought into closer adherence over time.

CaBLAM validation is not currently available through the phenix.refine GUI, but will be available on a future iteration of the validation tab.

Tools for exploring known structures are available through cablam.training

phenix.get_pdb_validation_report

get_pdb_validation_report.html

Description

This tool is for getting validation report from OneDep validation. Only mmCIF format is supported for models and diffraction data, and CCP4 format for EM maps. NMR chemical shifts and restraints are not currently supported. If validation completed successfully, a validation report in pdf and xml format will appear on hard drive.

It is not possible to predict how long will it take to get the report. Therefore after several attempts (see parameters to control duration and frequency of the requests), the tool will output a session id, which can be used later as a parameter to query RCSB again.

Note, running this tool again with files will create a new session on the validation server, not inquire about a previous job.

WARNING

This tool will send your model and data to RCSB servers for processing. Please refer to their privacy policy for details:

<https://www.wwpdb.org/about/privacy>

Command line usage examples

```
% phenix.get_pdb_validation_report model.cif % phenix.get_pdb_validation_report  
session_id=91c9cfa73036de43f744c7fa0f3b08c563bd9f9b % phenix.get_pdb_validation_report model.cif  
structure_factors.cif % phenix.get_pdb_validation_report model.cif map.ccp4
```

```
% phenix.get_pdb_validation_report model.cif
```

```
% phenix.get_pdb_validation_report session_id=91c9cfa73036de43f744c7fa0f3b08c563bd9f9b
```

```
% phenix.get_pdb_validation_report model.cif structure_factors.cif
```

```
% phenix.get_pdb_validation_report model.cif map.ccp4
```

GUI

The graphical user interface is currently available in the "PDB deposition" section. Also, the GUI will remember the session id; so when the job is restored, you should be able to query RCSB to see if the results are ready. When the results are fetched, the resulting page will change to show them.

List of all available keywords

timeout = 600 How long to wait for the result, in seconds
sping_frequency = 5 how often pool PDB for the result, in seconds
session_id = None Session ID for the job submitted previously
output_xml = True Fetch report in XML format
job_title = None Job title in PHENIX GUI, not used on command line
output_prefix = pdb_report
serial = None gui
GUI-specific parameter required for output directory
output_dir = None

- timeout = 600 How long to wait for the result, in seconds

- ping_frequency = 5 how often poll PDB for the result, in seconds
- session_id = None Session ID for the job submitted previously
- output_xml = True Fetch report in XML format
- job_title = None Job title in PHENIX GUI, not used on command line
- outputprefix = pdb_report serial = None
- prefix = pdb_report
- serial = None
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Evaluating a model with Holton geometry validation

holton_geometry_validation.html

Author(s)

- holton_geometry_validation: James Holton (Python coding by Tom Terwilliger)

Purpose

Summarize geometric quality of a model

How holton_geometry_validation works:

This tool summarizes the geometric quality of a model by calculating pseudo-energies (square of the value of the deviation from ideality divided by sigma) for every aspect of a model (each bond length, angle, rotamer, Ramachandran value, etc). For each metric (such as angles), the mean energy and worst energy are noted, and the probabilities of this mean and this worst value not occurring by chance are estimated. To reduce the dominance of very poor values, energy values E over 10 are filtered by reducing them toward 10 with a logarithmic function. The weighted energy for each metric is sum of the mean energy and the filtered worst energy for that metric, each weighted by their probabilities of not occurring by chance.

The overall energy is the sum of weighted energies for all metrics. The metrics used are:

```
CBetadev: Deviation of CB position from ideal (A)
Clash: Clash of non-bonded atoms (A)
Omega: Peptide omega angle deviation (degrees)
Rama: Ramachandran outlier probability
Rota: Rotamer outlier probability
Angle: Angle deviation (degrees)
Bond: Bond length deviation (A)
Chir: Chirality deviation (A**3)
Torsion: Torsion angle deviation (degrees)
Full-nonbond: Non-bonded deviations, including all instances
Nonbond: Non-bonded deviations (using Lennard-Jones potential,
    excluding bonds that are plausible and have low energies)
Plane: Planarity deviations (A)
```

Estimate of expected energy for ideal structure

This tool estimates the expected energy for a structure that has all bonds, angles, etc, distributed as normal distributions with the sigmas used in the geometry validation step above. That is, if the structure had a normal distribution of errors, it would have about this energy. The energy is estimated by sampling from a normal distribution when calculating the energy terms instead of taking the actual deviations. The sampling process is carried out many times (20) and the averages are reported.

Estimate of ratio of deviations to sigmas

An ideal model would have all individual energies distributed normally with variances approximately equal to the corresponding parent variances. The exact parent distributions for different structures would therefore be different. The distribution of overall scores for ideal models can be estimated by sampling all the contributions using normal distributions with appropriate parent variances.

The tool estimates the overall ratio of deviations to sigmas in three steps

First it lists all the values deviation/sigma for all the metrics. If the deviations were drawn from random Gaussian distributions with standard deviations of their corresponding sigmas, these values of deviation/sigma would be normally distributed with a standard deviation of one.

Then the 99.7th percentile of values of deviation/sigma is noted. For a Gaussian distribution, this percentile corresponds to 3 standard deviations. The estimated ratio of deviations/sigma overall is then 1/3 of this 99.7th percentile value, or 1 standard deviation of the ratios of deviation to sigma.

Comparing two models

If you want to be more certain about whether model A.pdb is better than B.pdb, you can supply both models to `holton_geometry_validation` (one as model and the other as `other_models`). The tool will calculate scores for A.pdb and B.pdb using not just the standard parameters, but all reasonable values of these parameters. The overall estimate of the difference in scores is obtained by averaging over the differences in scores using the sets of varied parameters. An overall difference and SD of the difference is supplied. If A.pdb consistently has a better score than B.pdb, you can be relatively confident that it really is better.

You can supply a list of models as well (or a model and a set of `other_models`), and each will be compared to the first model in the list.

Uses

You can use `holton_geometry_validation` to compare models for the same structure and identify which has fewer overall geometric problems.

You can use the expected energy to see if your structure has errors that are about what is expected based on ideal model geometry and uncertainties.

You can use the estimate of ratio of deviations to sigmas to further examine whether the deviations from ideality in your structure are about what is expected based on the expected variation in geometric values.

You can compare a series of models to see how each compares to the first model.

Examples

Standard run of `holton_geometry_validation`:

Running `holton_geometry_validation` is easy. From the command-line you can type:

```
phenix.holton_geometry_validation 1ss8_A.pdb
```

where 1ss8_A.pdb is the model you would like to evaluate.

Possible Problems

If your structure has hydrogen positions that do not match those expected by Phenix, you can ignore them in all metrics except non-bonded interactions with `ignore_h_except_in_nonbond=True`. Alternatively you can redo their positions with `keep_hydrogens=False`.

If you want to ignore just hydrogen positions in Arginine residues, you can specify `ignore_arg_h_nonbond=True`.

If you want to exclude hydrogen nonbond contacts involving waters, use `ignore_water_h_bonds`, and if you want to exclude all bonds involving hydrogen, `ignore_bond_lengths_with_h=True`

Limitations

The value of the score depends strongly on the worst energy in each category. This makes it possible to game the score (down to the next-worst values) by focusing attention on those individual energies. This limitation means that in the current implementation, refinement against the score is not useful. This limitation might be reduced by including not just the worst energy in each category, but rather weighted values of several or all the worst energies.

Parameters in the score function are somewhat arbitrary, and those that are not arbitrary do not have a perfect value. If parameters have ranges of reasonable values, the difference in score between two models cannot be known more accurately than the range of differences in score associated with reasonable values of the parameters.

Literature

The method is described at <https://bl831.als.lbl.gov/~jamesh/challenge/twoconf/#score>

Additional information

List of all available keywords

model = None input PDB file
other_models = None Models to compare with
model_get_individual_residue_scores = None Calculate individual residue scores in addition to overall
score_round_numbers = True Round numbers before
calculation
worst_clash_is_one_plus_n_times_worst_clash = True Scale worst clash score by (1 + n) where n
is total clashes
clash_energy_add_n = True Add number of clashes to clash energy
score_minimum_nonbond_score_to_be_worst = -0.1 Only include worst nonbond score if it is at least this
value
minimum_nonbond_score_to_be_included_in_average = 0 Only include nonbond score in average if it is
at least this value
include_full_nonbond_score = True If set, add additional scoring term in which all nonbond
values (even those less than the minimums) are included
keep_hydrogens = True If set, keep input hydrogens, but add any necessary riding H.
ignore_cis_peptides = False If set, ignore cis peptides. Otherwise (if False), penalize them.
ignore_h_except_in_nonbond = True If set, ignore H atoms except in nonbond
term
ignore_arg_h_nonbond = True If set, ignore H atoms in Arginine
ignore_bond_lengths_with_h = False If set, ignore bond lengths involving H atoms
ignore_water_h_bonds = False If set, ignore nonbonded contacts with water hydrogens
rotalyze_ramalyze_max_energy = 99 Maximum value of energy for a bad rotamer or phi-psi combination
overall_max_energy = None Maximum value of energy
omega_angle_sigma = 4 Sigma for omega angle
cbetadev_sigma = 0.05 Sigma for CB position
clashscore_ideal_dist = 3 Ideal distance in Lennard-Jones potential for clash
score_result
lj_dist_that_yields_zero = 6 Distance at which modified Lennard-Jones potential crosses zero
const_shrink_donor_acceptor = 0 Allow contacts closer by const_shrink_donor_acceptor from normal target for H-bonding atoms. NOTE: matches behavior of Phenix pre-2024.
remove_waters = None Remove waters before analysis
score_this_altloc_only = None Score only this altloc if specified
include_random = True Estimate expected value for structure with normal errors
n_random = 20 Number of samples for estimation of expected values
random_seed = 171927 Random seed
softPnna_params
y0 = 1 Parameter y0 for softPnna calculation
a2 = -0.0192266 Parameter a2 for softPnna calculation
a1 = 0.751694 Parameter a1 for softPnna calculation
a0 = 1.12482 Parameter a0 for softPnna calculation
mx = 0.21805 Parameter mx for softPnna calculation
my = 0.736621 Parameter my for softPnna calculation
variable_params
clashscore_ideal_dist_values = 2.5 3 3 3 3.5 Values to try for variables that are uncertain in comparison between models
lj_dist_that_yields_zero_values = 5 6 6 6 7
omega_angle_sigma_values = 4 5 6 7 8 9 10
cbetadev_sigma_values = .04 .05 .06

worst_clash_is_one_plus_n_times_worst_clash_values = 0 1 clash_energy_add_n_values = 0 1
 minimum_nonbond_score_to_be_worst_values = -0.1 0.0
 minimum_nonbond_score_to_be_included_in_average_values = 0 0 0 0 -1.1
 include_full_nonbond_score_values = 0 1 ignore_arg_h_nonbond_values = 0 1
 rotalyze_ramalyze_max_energy_values = 20 99 99 99 1000

- model = None input PDB file
- other_models = None Models to compare with model
- get_individual_residue_scores = None Calculate individual residue scores in addition to overall score
- round_numbers = True Round numbers before calculation
- worst_clash_is_one_plus_n_times_worst_clash = True Scale worst clash score by (1 + n) where n is total clashes
- clash_energy_add_n = True Add number of clashes to clash energy score
- minimum_nonbond_score_to_be_worst = -0.1 Only include worst nonbond score if it is at least this value
- minimum_nonbond_score_to_be_included_in_average = 0 Only include nonbond score in average if it is at least this value
- include_full_nonbond_score = True If set, add additional scoring term in which all nonbond values (even those less than the minimums) are included
- keep_hydrogens = True If set, keep input hydrogens, but add any necessary riding H.
- ignore_cis_peptides = False If set, ignore cis peptides. Otherwise (if False), penalize them.
- ignore_h_except_in_nonbond = True If set, ignore H atoms except in nonbond term
- ignore_arg_h_nonbond = True If set, ignore H atoms in Arginine
- ignore_bond_lengths_with_h = False If set, ignore bond lengths involving H atoms
- ignore_water_h_bonds = False If set, ignore nonbonded contacts with water hydrogens
- rotalyze_ramalyze_max_energy = 99 Maximum value of energy for a bad rotamer or phi-psi combination
- overall_max_energy = None Maximum value of energy
- omega_angle_sigma = 4 Sigma for omega angle
- cbetadev_sigma = 0.05 Sigma for CB position
- clashscore_ideal_dist = 3 Ideal distance in Lennard-Jones potential for clashscore result
- lj_dist_that_yields_zero = 6 Distance at which modified Lennard-Jones potential crosses zero
- const_shrink_donor_acceptor = 0 Allow contacts closer by const_shrink_donor_acceptor from normal target for H-bonding atoms. NOTE: matches behavior of Phenix pre-2024.
- remove_waters = None Remove waters before analysis
- score_this_altloc_only = None Score only this altloc if specified
- include_random = True Estimate expected value for structure with normal errors
- n_random = 20 Number of samples for estimation of expected values
- random_seed = 171927 Random seed
- softPnna_paramsy0 = 1 Parameter y0 for softPnna calculation
 a2 = -0.0192266 Parameter a2 for softPnna calculation
 a1 = 0.751694 Parameter a1 for softPnna calculation
 a0 = 1.12482 Parameter a0 for softPnna calculation

softPnna calculationmx = 0.21805 Parameter mx for softPnna calculationmy = 0.736621 Parameter my for softPnna calculation

- y0 = 1 Parameter y0 for softPnna calculation
- a2 = -0.0192266 Parameter a2 for softPnna calculation
- a1 = 0.751694 Parameter a1 for softPnna calculation
- a0 = 1.12482 Parameter a0 for softPnna calculation
- mx = 0.21805 Parameter mx for softPnna calculation
- my = 0.736621 Parameter my for softPnna calculation
- variable_paramsclashscore_ideal_dist_values = 2.5 3 3 3 3.5 Values to try for variables that are uncertain in comparison between modelslj_dist_that_yields_zero_values = 5 6 6 6 7
omega_angle_sigma_values = 4 5 6 7 8 9 10 cbetadev_sigma_values = .04 .05 .06
worst_clash_is_one_plus_n_times_worst_clash_values = 0 1 clash_energy_add_n_values = 0 1
minimum_nonbond_score_to_be_worst_values = -0.1 0.0
minimum_nonbond_score_to_be_included_in_average_values = 0 0 0 0 -1.1
include_full_nonbond_score_values = 0 1 ignore_arg_h_nonbond_values = 0 1
rotalyze_ramalyze_max_energy_values = 20 99 99 99 1000
- clashscore_ideal_dist_values = 2.5 3 3 3 3.5 Values to try for variables that are uncertain in comparison between models
- lj_dist_that_yields_zero_values = 5 6 6 6 7
- omega_angle_sigma_values = 4 5 6 7 8 9 10
- cbetadev_sigma_values = .04 .05 .06
- worst_clash_is_one_plus_n_times_worst_clash_values = 0 1
- clash_energy_add_n_values = 0 1
- minimum_nonbond_score_to_be_worst_values = -0.1 0.0
- minimum_nonbond_score_to_be_included_in_average_values = 0 0 0 0 -1.1
- include_full_nonbond_score_values = 0 1
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MolProbity - An Active Validation Tool

molprobity_tool.html

Contents

- Authors
- Purpose
- Usage
- Possible Problems
- Literature

Authors

MolProbity is a web application that integrates validation programs from the Richardson lab at Duke University.

- Ian Davis, principal author: PHP/Java web service; KiNG; Ramachandran & Rotamer; Dangle
- Vincent Chen: extensions to KiNG & MolProbity
- Mike Word: Reduce; Probe; Clashlist
- Dave Richardson: kinemages; Mage; Prekin; Suitename
- Xueyi Wang: RNABC
- Jack Snoeyink & Andrew Leaver-Fay: Reduce update
- Bryan Arendall: webmaster; databases

Purpose

MolProbity provides the user with an expert-system consultation about the accuracy of a macromolecular structure model, diagnosing local problems and enabling their correction. It combines all atom contact analysis with updated versions of more traditional tools for validating geometry and dihedral-angle combinations. MolProbity is most complete for crystal structures of proteins and RNA, but also handles DNA, ligands, and NMR ensembles. It works best as an active validation tool - used as soon as a model is available and during each rebuild/refine loop, not just at the end to provide global statistics before deposition. It produces coordinates, graphics, and numerical evaluations that integrate with either manual or automated use in systems such as PHENIX, KiNG, or Coot.

Usage

The integrated MolProbity web application is at <http://molprobity.biochem.duke.edu/>. The user is guided through a work-flow that typically consists of:

- Fetch or upload model(s)
- Add & optimize H atoms, with correction of Asn/Gln/His flips
- Calculate per-residue & global quality analyses: all-atom steric clashes geometry (e.g., Cbeta or ribose pucker ideality) Ramachandran, sidechain rotamer, or RNA backbone outliers global MolProbity score

Calculate per-residue & global quality analyses:

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- all-atom steric clashes
- geometry (e.g., Cbeta or ribose pucker ideality)
- Ramachandran, sidechain rotamer, or RNA backbone outliers
- global MolProbity score
- View multi-criterion chart and/or on-line 3D KiNG graphics summaries
- [Optional features, e.g. interface analysis; load maps for on-line viewing; Coot to-do list]
- Download coordinate & graphics files for further work on local corrections

An increasingly broad subset of MolProbity functionalities are integrated directly into PHENIX for use in refinement, Resolve, and wizard decisions. These are all available in the PHENIX GUI as part of the `comprehensive validation interface`, which runs all available analyses and provides convenient links to molecular graphics software. Phenix.reduce provides optimized hydrogen addition, phenix.probe and quick_clashlist.py provide all-atom clash analysis, and python versions of the Ramachandran and rotamer scores are available in the mmtbx. Interactive all-atom contact dots are also available in Coot.

Possible Problems

Web usage requires Java, Javascript, and a modern web browser.

MolProbity provides reasonable session protection, but if security or large-scale usage are at issue, you can install MP to run on your own Linux or Mac computer provided that the computer has a web server (Apache), the PHP scripting language, JAVA, and a few common Unix utility programs. For more information, follow the "Download MolProbity" link on the MP main page.

Literature

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Crystallographic Validation tools in Phenix

validation.html

Contents

- Overview
- Running validation
- MolProbity summary
- Geometry restraints outliers
- Validation of protein geometry
- All-atom contacts
- RNA validation
- Real-space correlation
- POLYGON
- Suggestions for improving models
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Overview

This document covers the use of the validation software in the Phenix GUI, which is run both as a standalone program and automatically as part of phenix.refine. Much of this software is derived from the MolProbity web server. Although it is possible to run validation with only a model, we recommend including experimental data, as interpreting geometry outliers only makes sense in the context of the electron density maps.

For running with crystallographic data (X-ray or neutron), use the Comprehensive Validation GUI (X-ray). For running with cryo-EM data, use the Comprehensive Validation GUI (Cryo-EM) (see [validation_cryo_em](#)).

The analyses performed by Phenix include:

identification of outliers from restrained geometric parameters (bonds, angles, etc.) Ramachandran plot and outlier info identification of disfavored protein sidechain rotamers (and chi1-chi2 plot) all-atom contact analysis with explicit hydrogens (including clashscore) suggestions for asymmetric sidechain flips (Asn, Gln, His) real-space correlation to electron density (RSCC) multi-criterion plot of B, RSCC, density values, and MolProbity outliers by residue numbers

- identification of outliers from restrained geometric parameters (bonds, angles, etc.)
- Ramachandran plot and outlier info
- identification of disfavored protein sidechain rotamers (and chi1-chi2 plot)
- all-atom contact analysis with explicit hydrogens (including clashscore)
- suggestions for asymmetric sidechain flips (Asn, Gln, His)
- real-space correlation to electron density (RSCC)
- multi-criterion plot of B, RSCC, density values, and MolProbity outliers by residue numbers

Any non-protein molecule will be included in the real-space fit evaluation, and if CIF files are provided, geometric outliers will be detected as well. Comprehensive validation of nucleic acid structures is planned for a future release.

All of the outliers listed in the GUI link to supported molecular graphics programs (currently Coot, PyMOL, and the internal viewer) if launched from within Phenix. Clicking an outlier will recenter the graphics window on that atom or residue, and in some cases will also highlight the relevant selection. If experimental data are included, buttons in the result window will open the model and maps in the graphics programs.

Running validation

If you are performing crystallographic refinement with the `*phenix.refine*` GUI, or real-space refinement with the `*phenix.real_space_refine*` GUI, the validation will be performed automatically as soon as the final model is ready. To validate a model with diffraction data, launch the "Comprehensive validation" app from the main GUI. At a minimum, you will need a PDB file, and should probably add a reflections file containing intensities or amplitudes and R-free flags, plus any CIF files required by non-standard ligands in the structure. Most parameters should be extracted automatically.

After running the validation, a new window will open with the results. If experimental data were included, the first tab will summarize the statistics including R-factors, plus data plots. Buttons for Coot and PyMOL will also appear. Additional tabs contain the MolProbity validation suite, sanity checks for atomic properties (B-factor and occupancy), and real-space correlation.

MolProbity summary

An initial tab shows the target and actual values for assorted validation criteria (mostly specific to proteins). See below for advice on interpreting these results. Residues with missing atoms are listed in a separate section; although these are not necessarily errors, they are often unintentionally left incomplete.

Geometry restraints outliers

Phenix, like most crystallographic software, uses the Engh and Huber (1991) restraints for proteins, nucleic acids, and other common molecules, here in the form of the CCP4 monomer library. Other restraints come from CIF files provided by the user; these can be generated by `phenix.elbow` and associated programs. These restraints are used in refinement to prevent distortions of model geometry, and to increase the observation-to-parameter ratio. The default restraints are for bond lengths, bond angles, dihedral (torsion) angles, chiral centers, planar groups (such as aromatic rings), and nonbonded (VDW) interactions. All of these are analyzed here except for the nonbonded restraint outliers, which are made redundant by the much more thorough all-atom contact analysis (see below).

All restrained values have an associated "sigma", i.e. the observed standard deviation from that value in very high-resolution structures; this is used to weight the restraint during refinement. Large deviations (greater than 4 sigma) from the restrained values usually indicate something seriously wrong with the model, and should not be taken as examples of genuine structural strain unless the electron density for that feature is exceptionally good (usually only at ultra-high-resolution, i.e. better than 1.0 Angstrom) and the deviation is no more than approximately 7 sigma. At moderate-to-low resolutions, there should be almost no outliers, although strained dihedral angles are possible.

Validation of protein geometry

The geometry of protein chains is restricted by additional empirical observations that are not part of the standard restraints. The "Protein" tab contains these analyses. Many structures contain residues that have been correctly placed but are flagged as Ramachandran or rotamer outliers; however, there are rarely more than a handful of these. At resolutions below 3.0Å, any outliers should be considered errors. No C-beta

outliers are acceptable at anything worse than sub-atomic ($< 0.8\text{\AA}$) resolution.

The rest of the tab is divided into several sections:

The Ramachandran plot, sets limits on the combination of Phi (C-N-CA-C) and Psi (N-CA-C-N) dihedral angles for any residue. In Phenix and MolProbity, the standard distribution of values is taken from the Top8000 database of high-resolution protein structures. The main window contains a list of scored outliers (the lower the score, the worse the residue), and all residues are plotted graphically in a separate window. The scoring depends on the residue type and relative position; Gly, Pro, and adjacent residues depend on different distributions, all of which can be viewed in the plot window. Rotamer comparisons also use the Top8000 database, and all sidechain Chi angles for a residue are evaluated together. Any residue which has an outlying sidechain dihedral restraint is likely to be flagged by this analysis too. The distribution of Chi1-Chi2 angle combinations for each residue type is also plotted in a separate window, against a background distribution taken from the Top8000 database. (Note that longer residues may be flagged as outliers based on the value of Chi3 or Chi4, even though they fall into the favored region of the Chi1-Chi2 plot.) C-beta atoms deviating from the proper position are flagged; these are caused by incompatible sidechain and mainchain positions, which are usually due to incorrect fitting to the density. This will most likely be reflected in the comparison to restraints, with significant deviation from chirality and bond lengths or angles.

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All-atom contacts

Nonbonded restraints used in refinement can function with or without explicit hydrogen atoms; however, if hydrogens are absent, the atomic radii of other atoms will be increased to compensate. This approximation works decently on a global structural level, but often leaves chemically impossible geometries in place. Therefore, for this step, hydrogens are added to the model (proteins and nucleic acids only at this time) by phenix.reduce; this will first strip off any existing hydrogens. Reduce will flag residues whose sidechains require flipping based on hydrogen-bonding geometry and clashes caused by newly added hydrogens. These include Asn, Gln, and His, which are easily mis-fit due to the apparent symmetry of the sidechain without hydrogens.

Once hydrogens have been added, phenix.probe is run to analyze the atomic contacts. All atomic overlaps worse than $0.4\text{\AA}$ are listed, and a global "clash score" is reported for the entire structure (calculated as $1000 * (\text{number of bad overlaps}) / (\text{number of atoms})$). This should be as low as possible, although a value of zero is

both difficult and unusual. We strongly recommend refining with explicit hydrogens at any resolution, as this typically improves overall model geometry (and rarely makes anything worse).

Since Coot is able to read the output of Probe and graphically display the contacts as dots and dashes, this information will be automatically read into Coot when run from Phenix. Display of the dots can be toggled using the external validation window popped up in Coot. Bad overlaps will be drawn as hot-pink dashes. It is usually much easier to visualize the cause of the problem if hydrogens are drawn as well.

RNA validation

A more robust set of target bond and angle values for RNA chains is applied separately from the parameters in the CCP4 monomer library. This includes conformation-dependent restraints on the backbone sugar pucker, illustrated below. Improvement of geometric refinement of RNA structures is an ongoing project with PHENIX; currently, most RNA structures exhibit at least a few outliers (and sometimes many).

Real-space correlation

This is the only part of validation that requires the underlying diffraction data. Phenix will perform bulk-solvent correction and scaling on the data and calculate a likelihood-weighted 2mFo-DFc map. This is compared to a map calculated from the model alone, and correlation coefficients for each residue are obtained. At resolutions better than 2.5 Angstroms, the values for individual atoms will also be displayed. In the GUI, these lists can be filtered and/or sorted by CC.

Although real-space CC can be a useful indicator of poorly fit regions of a molecule, it should not be interpreted as an absolute score, and it is difficult to identify an ideal cutoff below which the score should never fall. However, it is generally useful to inspect the 2mFo-DFc and mFo-DFc maps for residues which score below 0.8. Some of these will be correctly placed but poorly ordered; this is not a problem as long as the geometry is within normal bounds. The B-factor, occupancy, and absolute density values should also be considered when evaluating the results, as these may indicate parts of the structure that are intrinsically disordered. The multi-criterion plot displays these values (over 100-residue regions) along with validation outliers. Clicking anywhere inside the plot will zoom in on the corresponding residue at that position in Coot or PyMOL.

POLYGON

The program POLYGON (Urzhumtseva et al. 2009) has been ported to the Phenix GUI for comparing model quality indicators to similar structures in the PDB. Pre-computed values for a selection of 1000 structures determined at a similar resolution are plotted radially as one-dimensional histograms. The score for the model for each of these statistics is also plotted on the histograms, and the lines connecting these points form a complete polygon. For a high-quality, well-refined structure, the shape should be approximately symmetric and relatively small.

Suggestions for improving models

Rotamer fixing and N/Q/H sidechain flips are now options in phenix.refine, the latter now part of the default settings. There are currently no fully automatic corrections for any of the other problems identified in validation. phenix.autobuild can be used to rebuild problem regions of the structure, but this becomes more difficult as the quality of the map decreases or the resolution gets worse. Re-refinement with a slightly different protocol is often helpful; in particular, explicit hydrogens can help constrain the model. The weight applied to X-ray terms during refinement may need to be reduced in favor of geometric restraints; this can be

done automatically by phenix.refine.

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